

Visual Photonics Epitaxy



Initiation: Transitioning to photonics

Key Points

- VPEC, as an upstream supplier for transceiver production, is positioned to ride an AI networking boom, amplified by tight supply of laser chips.
- With surging Photonics revenue, we view it as becoming more of an optical components play, yet, for now, it trades more like a microelectronics name.
- Skyworks/Qorvo merger impact limited as both are VPEC customers and as competition fades. Initiate at Outperform on a NT\$236 TP (30x 27E PE).

- We expect the rise of AI scale-out networking to translate to surging transceiver demand, forecasting a 70% market revenue CAGR over 2024-27E. VPEC, as a supplier of epiwafers for optical chips, such as laser diodes and photodetectors, working with GCS (4991 TT, Not Rated) and also a Japanese laser vendor, has significant exposure to the AI scale-across market. We expect the firm's Photonics business growth to hit 40% YoY in 2026E and 46% in 2027E, repositioning it as primarily an optical communications company.
- We anticipate tighter supply of laser chips in 2026E, as we believe the industry is unprepared for explosive demand from AI clusters and scale-out networking.
- We believe supply shortages will initially emerge upstream, creating opportunities for new entrants and expect VPEC, with its broad customer base - including TW, JP and US customers - to pick up a larger proportion of orders. We forecast a 91% revenue CAGR for VPEC's data-centre photonics business over 2026/27E, and see gross margin rising 11ppts over two years to hit ~48% by 2027E.
- For its microelectronics division, we see impact from Skyworks and Qorvo's planned merger as limited, as both companies are already VPEC customers, and, as VPEC notes, the two companies will still continue to operate independently in 2026E. We believe VPEC will sustain its margin as competition from major rival IQE (NR) eases given its plan to sell its Taiwan business.

Valuation: Our NT\$236 target price is based on 30x 2027E EPS, referencing its average since 2018 and supported by our 64% two-year EPS CAGR; we see upside as the market recognises its repositioning as an optical communications company.

Catalysts: GB300 shipment, CW laser qualification from key customer

Investment Thesis and Recommendation

Initiate at Outperform. We believe VPEC will be a major beneficiary from spill-over demand for optical chips as a result of which, it will transform into an optical/photonics company.

Semicons & Semicon Equip Taiwan



Nick Hung



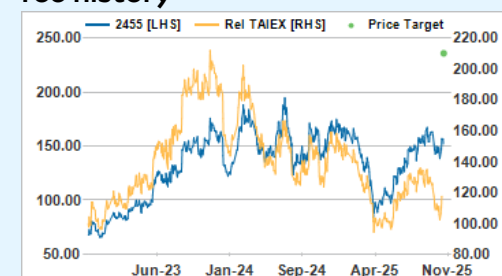
Arthur Lai

2455 TT	Outperform
Price (at 13 Nov 2025)	TWD156.50
12-month target	TWD236.00
12-month TSR (%)	53.4
Volatility Index	High
Market Cap (Local) (m)	28,938
Market Cap (USD) (m)	932
Free Float (%)	94
30-day avg turnover (USD) (m)	25.6

Investment Fundamentals

Year end 31 Dec	2024A	2025E	2026E	2027E
Revenue (m)	3,241	3,378	4,630	5,373
Revenue growth (%)	20.3	4.2	37.1	16.1
EBIT (m)	721.2	670.7	1,245	1,721
EBIT growth (%)	33.0	(7.0)	85.6	38.3
Reported profit (m)	671.1	543.6	1,081	1,462
Adjusted profit (m)	671.1	543.6	1,081	1,462
EPS rep [TWD]	3.6	2.9	5.8	7.9
EPS rep growth (%)	49.0	(19.2)	98.0	35.3
EPS adj [TWD]	3.6	2.9	5.8	7.9
EPS adj growth (%)	49.0	(19.3)	98.3	35.3
Net debt/equity (%)	(20.3)	(13.1)	(11.9)	(23.1)
ROA (%)	15.7	14.5	25.4	31.4
ROE (%)	21.0	16.4	31.2	36.6
PER rep (x)	43.1	53.4	26.9	19.9
PER adj (x)	43.1	53.4	26.9	19.9
EV/EBITDA (x)	27.6	29.1	18.1	13.7
P/BV (x)	8.7	8.8	8.0	6.7
Total div yield (%)	2.1	2.6	2.6	2.6
Quant (rank vs. global sector)	425 / 650			

2455 TT rel TAIEX performance, & rec history



Source: FactSet, Macquarie Research, Nov 2025 (all figures in TWD unless noted, TP in TWD)

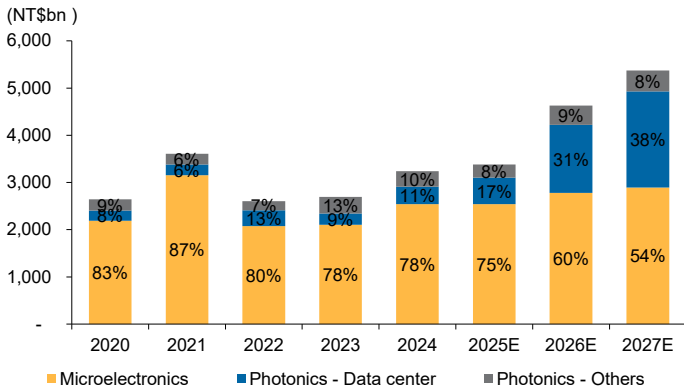
Key Risks to Investment Thesis

- Slower-than-expected transceiver demand.
- Over-capacity of leading players
- Declining wireless communication market

Company Description

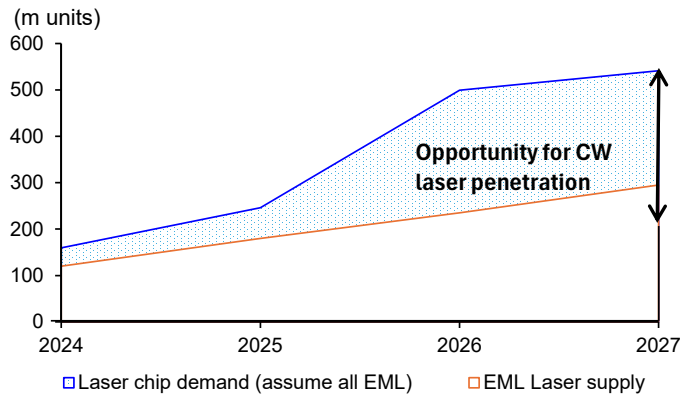
VPEC is a global leader in compound semiconductors, specialising in GaAs and InP epiwafers, the upstream raw materials for several communication components, such as power amplifiers (PA) in wireless communications and transceivers in data centre interconnect. The company has a dominant share of the wireless PA epiwafer market, and it is expanding its position in the optical components market for transceivers.

Figure 1 - Revenue breakdown



Source: Company, Macquarie Research, Nov 2025

Figure 2 - EML shortage gives way to CW laser penetration



Source: Macquarie Research, Nov 2025

Visual Photonics Epitaxy (2455 TT) TWD/(m) unless otherwise noted

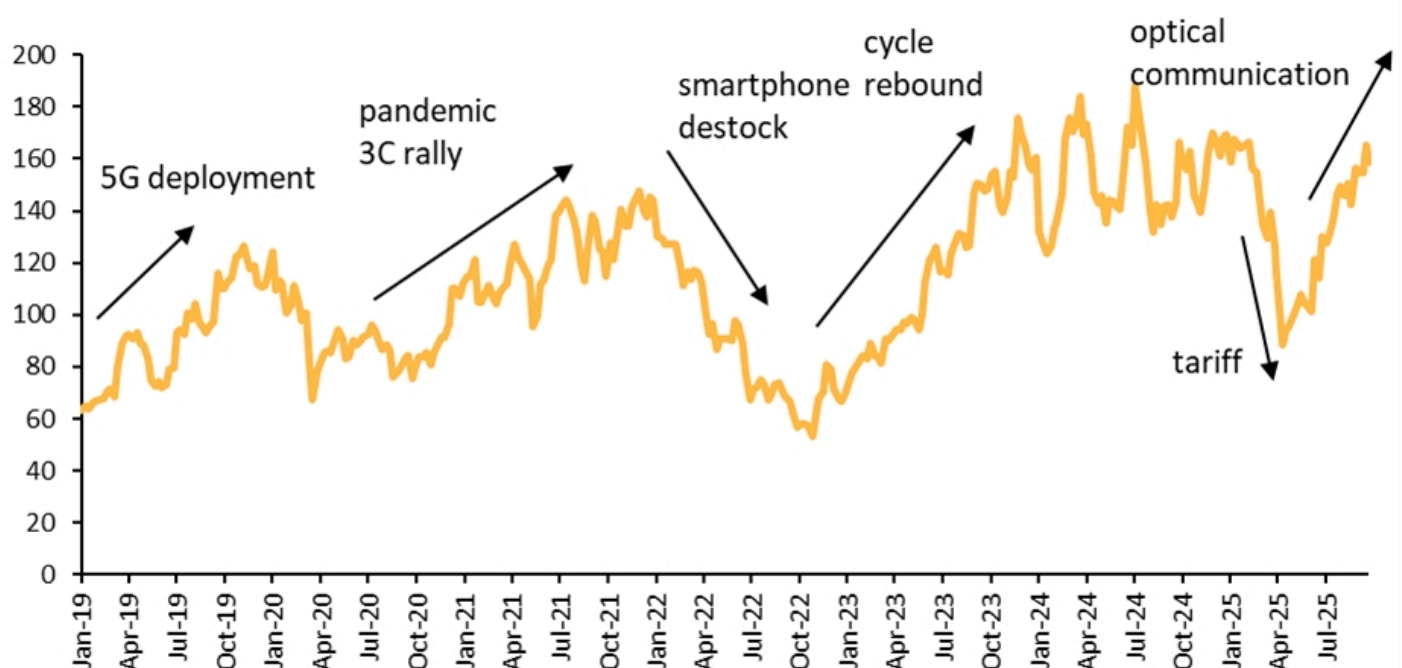
Income Statement	Dec FY	2024A	2025E	2026E	2027E	Q3/25A	Q4/25E	Balance Sheet	2024A	2025E	2026E	2027E
Revenue		3,241	3,378	4,630	5,373	912.8	957.9	Cash	1,176	1,032	1,033	1,603
Cost of Goods Sold		1,962	2,147	2,658	2,808	612.9	588.6	Receivables	380.7	504.7	668.2	731.4
Gross Profit		1,279	1,231	1,972	2,565	299.9	369.2	Inventories	618.6	787.6	954.7	964.6
EBITDA		1,009	961.8	1,548	2,042	232.2	283.8	Fixed Assets	2,262	2,104	2,204	2,286
Depreciation		286.0	288.3	299.8	317.7	72.3	72.6	Intangibles	0.0	0.0	0.0	0.0
Amortisation		2.2	2.8	2.8	2.8	0.7	0.7	Other Assets	146.1	257.4	254.5	251.7
EBIT		721.2	670.7	1,245	1,721	159.2	210.5	Total Assets	4,583	4,685	5,114	5,837
Net Interest Income		13.6	8.1	1.4	1.4	0.4	0.4	Payables	336.9	428.9	519.9	525.3
Associates		0.0	0.0	0.0	0.0	0.0	0.0	Short Term Debt	0.0	0.0	0.0	0.0
Exceptionals		0.0	0.0	0.0	0.0	0.0	0.0	Long Term Debt	500.0	600.0	600.0	600.0
Forex Gains / Losses		73.2	(72.2)	0.0	0.0	15.8	0.0	Other Liabilities	416.7	361.8	361.8	361.8
Other Pre-Tax Income		9.7	51.6	104.2	104.2	25.1	26.1	Total Liabilities	1,254	1,391	1,482	1,487
Pre-Tax Profit		817.7	658.3	1,351	1,827	200.4	236.9	Shareholders' Funds	3,372	3,337	3,675	4,392
Tax Expense		(146.6)	(114.7)	(270.1)	(365.4)	(39.9)	(47.4)	Minority Interests	0.0	0.0	0.0	0.0
Net Profit		671.1	543.6	1,081	1,462	160.5	189.5	Other	(42.3)	(42.3)	(42.3)	(42.3)
Minority Interests		0.0	0.0	0.0	0.0	0.0	0.0	Total S/H Equity	3,329	3,294	3,632	4,350
Reported Earnings		671.1	543.6	1,081	1,462	160.5	189.5	Total Liab & S/H Funds	4,583	4,685	5,114	5,837
Adjusted Earnings		671.1	543.6	1,081	1,462	160.5	189.5	Net Debt / Equity (%)	(20.3)	(13.1)	(11.9)	(23.1)
Basic Shares Outstanding		184.9	186.0	186.0	186.0	186.0	186.0	Interest Cover (x)	NM	NM	NM	NM
Diluted Shares Outstanding		184.9	185.3	186.0	186.0	186.0	186.0	ROE (%)	21.0	16.4	31.2	36.6
EPS (rep) [TWD]		3.6	2.9	5.8	7.9	0.9	1.0	ROA (%)	15.7	14.5	25.4	31.4
EPS (adj) [TWD]		3.6	2.9	5.8	7.9	0.9	1.0	ROIC (%)	19.4	20.9	34.8	43.0
Total DPS [TWD]		3.2	4.0	4.0	4.0	4.0	0.0					
Ratio		2024A	2025E	2026E	2027E			Cash Flow Analysis	2024A	2025E	2026E	2027E
Revenue Growth (%)		20.3	4.2	37.1	16.1	-	-	EBITDA	1,009	961.8	1,548	2,042
EBITDA Growth (%)		22.2	(4.7)	60.9	31.9	-	-	Tax Paid	(146.6)	(114.7)	(270.1)	(365.4)
EBIT Growth (%)		33.0	(7.0)	85.6	38.3	-	-	Chgs in Working Cap	68.0	(186.5)	(239.6)	(67.7)
EPS Growth (adj) (%)		49.0	(19.3)	98.3	35.3	-	-	Net Interest Paid	13.6	8.1	1.4	1.4
Gross Profit Margin (%)		39.5	36.4	42.6	47.7	-	-	Other	141.4	(96.4)	104.2	104.2
EBITDA Margin (%)		31.1	28.5	33.4	38.0	-	-	Operating Cashflow	1,086	572.2	1,144	1,714
EBIT Margin (%)		22.3	19.9	26.9	32.0	-	-	Acquisitions	0.0	0.0	0.0	0.0
Net Profit Margin (%)		20.7	16.1	23.3	27.2	-	-	Capex	(43.8)	(198.7)	(400.0)	(400.0)
Payout Ratio (%)		88.7	136.6	68.9	50.9	-	-	Other	(3.0)	(2.5)	0.0	0.0
PE (rep) (x)		43.1	53.4	26.9	19.9	-	-	Investing Cashflow	(46.7)	(201.2)	(400.0)	(400.0)
PE (adj) (x)		43.1	53.4	26.9	19.9	-	-	Dividend (Ordinary)	(406.8)	(591.7)	(742.3)	(744.1)
EV/EBITDA (x)		27.6	29.1	18.1	13.7	-	-	Debt Movements	1,800	600.0	0.0	0.0
EV/EBIT (x)		38.6	41.8	22.5	16.3	-	-	Other	(2,104)	(502.9)	0.0	0.0
Price/Book (x)		8.7	8.8	8.0	6.7	-	-	Financing Cashflow	(710.7)	(494.6)	(742.3)	(744.1)
Total Div Yield (%)		2.1	2.6	2.6	2.6	-	-	Net Chg in Cash/Debt	350.0	(144.1)	1.2	570.2
								Free Cashflow	1,042	373.5	743.6	1,314

Source: Company data, Macquarie Research Nov 2025

Investment thesis

- We expect exponential demand growth for AI networking, driven by larger clusters. We forecast a 50% networking switch market CAGR and 70% transceiver market CAGR over 2024-27E. As clusters scale, the increased layers of interconnect will drive multiples more demand for networking switches and transceivers.
- We have already noted tightening supply with comments from downstream transceiver makers on EML shortages as early as 2024. We expect the shortfall to widen in 2026E, which we believe will result in greater adoption of continuous wave (CW) lasers for use in silicon photonic (SiPh) transceivers as an alternative to EML in transceivers. As a more open market than EML, we expect the shift to CW lasers will allow new entrants to enter the upstream supply chain.
- We expect VPEC to benefit from spill-over orders for optical chips. VPEC entered the CW laser market supplying a Japanese optical chip supplier for Innolight. It has also built deep collaboration with GCS, the world's largest outsourced photodetector (PD) producer, leveraging 4-inch indium phosphide (InP) capacity to achieve cost and efficiency advantages. We believe VPEC will continue to leverage its broad clientele from its Microelectronics business to expand its Photonics business, given the crossover in its product offerings across the two divisions.
- In its Microelectronics business, its main competitor, IQE, has announced plans to sell its Taiwan subsidiary, suggesting the competitive landscape for VPEC may ease. Even if we take a conservative view on the smartphone market, we still believe VPEC is well positioned in the wireless communication market. Its Microelectronics business offers a measure of downside protection to the more volatile earnings for the Photonics business as well as steady cashflow, as it maintains its leading position in the gallium arsenide (GaAs) market.
- We believe the recent Skyworks/Qorvo merger to form a semis giant will have limited impact throughout 2026 as the 2 companies will work separately under regulatory requirements.
- We believe a rerating is possible for VPEC, as its high-margin photonic business will reach 40% of total sales in 2026E and potentially over 46% in 2027E. We believe the spill-over effect will further accelerate the process, leading to potential upside for VPEC.

Figure 3 - VPEC stock price event chart

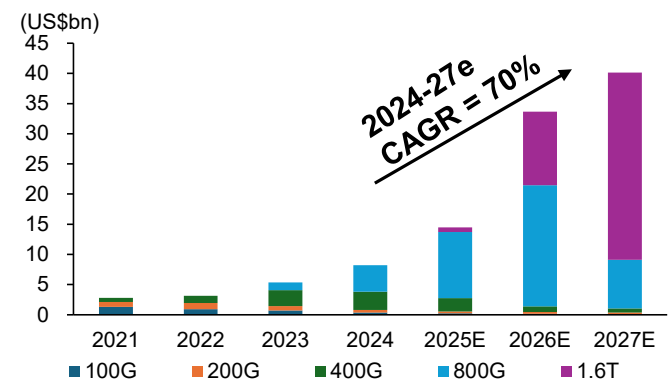


Source: Bloomberg, Macquarie Research, Nov 2025

Transceiver market to explode starting 2H25

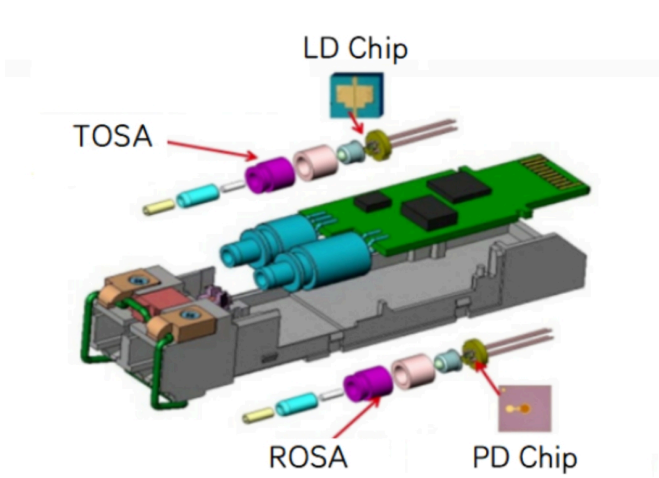
- We forecast a 70% transceiver market CAGR over 2024-27E, with growth skewed towards 2026-27E.
- Transceivers require optical chips, including laser diodes (LD) and photodetector s. VPEC manufactures epi-wafers, the raw material used in optical chip production.
- We expect VPEC's Photonics division to deliver 48% HoH revenue growth in 2H25E and 121% YoY in 2026E, as we believe VPEC will benefit from outsourcing demand for transceiver PDs from major optical players, as demand explodes in 2026E.
- VPEC has a long history of manufacturing epiwafers for transceiver PD components. Its customer, GCS, is the world's leading supplier of outsourced PDs, with greater market share in 4" InP process. We expect the outsourced opportunity to continue to boom as capacity remains tight across the industry.

Figure 4 - MQ transceiver market forecast



Source: IDC, Macquarie Research, Nov 2025

Figure 5 - Transceiver component overview



Source: Fibermall, Nov 2025

Figure 6 - Transceiver supply chain overview

Transceiver Component	Laser Diode (Laser chip)	Modulator	Photodetector	PCB	Passive Components	DSP
Transceiver: Innolight Eoptolink Fabrinet Coherent Lumentum Accelink AAOI Ciena TOSA/ROSA TFC Elaser Luxnet Truelight EZConn	EML Laser Lumentum Coherent Sumitomo Broadcom Mitsubishi CW Laser EML players Sumitomo Landmark Yuanjie Furukawa AAOI	SiPh Modulator TSMC UMC Global Foundry Tower semiconductor	Photodetector EML/CW players GCS Dexerials Epi-wafer Landmark VPEC	HDI SCC Unimicron ZDT Compeq	Fiber /connector: TFC Senko Fabrinet FOCI Browave Sumitomo Furukawa Corning Fujikura EZconn	DSP Broadcom Marvell MaxLinear Airoha

Source: Macquarie Research, Nov 2025

Beneficiary of laser shortage

- As a laser epiwafer producer, we believe VPEC is one of the firms most exposed to industry spillover demand for CW lasers. We expect significant growth in its CW laser business in 2026E, starting off from a low base and accelerating with what we anticipate will be exponential demand growth for data centre interconnect components.
- VPEC focuses on epiwafer and related subsequent processes. Its long-term experience with MOCVD (metal-organic chemical vapour deposition) underpins its upstream manufacturing processes for laser chips (Figure 7).

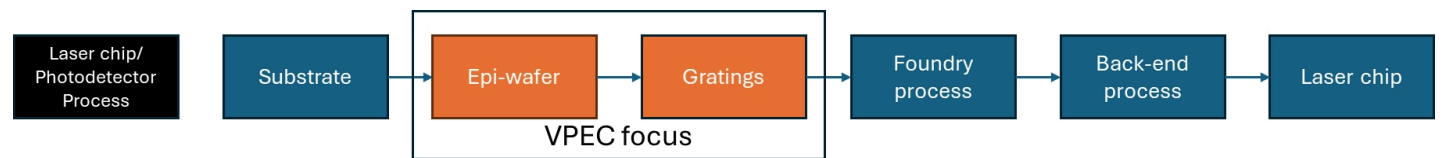
Q: Why there is a laser industry shortage, and how do we expect it to play out?

- We see evidence of a shortage of electro-absorption modulated lasers (EML, used in high-end transceivers) since 2024, from comments across the industry in earnings calls. The shortage has been driven primarily by surging demand for optical interconnect for high-speed networking.
- As we lay out in our **AI networking** Ecosystem report, we believe the networking industry will exhibit rapid growth in 2026-27E, creating a non-cyclical demand uptick for components, resulting in a demand shock for the unprepared supply chain. The laser industry has been loss-making for a long time, even for leading players, and thus there has been a reluctance across the industry to expand capacity.
- We forecast demand for ~500mn EML chips in 2026E, assuming all transceivers use EML as the light source. However, our channel checks suggest only ~235mn EML chips can be produced. The EML shortage started in 2024E, but leading EML players' capacity expansion plans have been conservative, extending the shortage.
- Thus, we expect the transceiver industry to shift away from EML and towards new SiPh solutions for their transceivers, as these only require CW lasers to function, allowing for more suppliers to compete.

Q: Why do we think VPEC will be a major beneficiary?

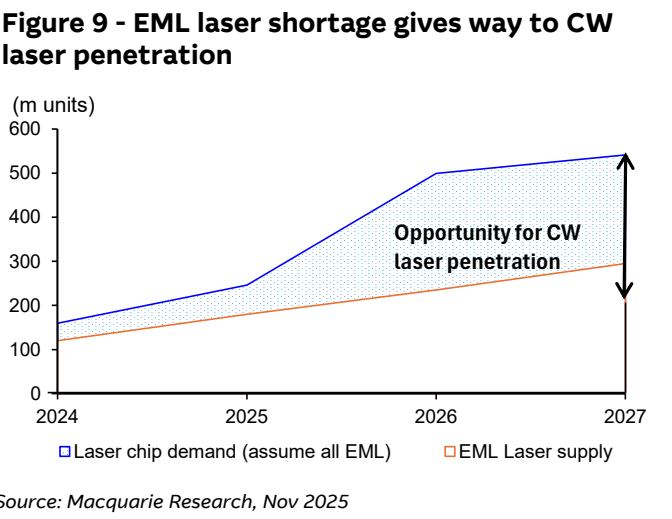
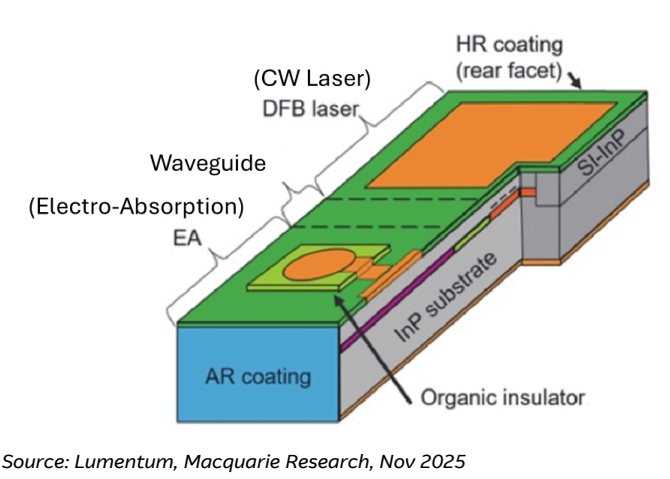
- We believe VPEC has already secured enough InP substrate for manufacturing and that it started shipping its 3" InP laser epiwafer in late 2024.
- VPEC has also worked with GCS on CW lasers, which we believe will invite more demand for VPEC's 3" inP wafer in 2026E. VPEC boasts 60+ MOCVD reactors, which is more than enough to cater to spill-over orders from industry leaders.
- We assume VPEC will obtain more outsourced PD orders for epiwafers from a key US laser player, which has been a Microelectronics business customer, as, given the tight market for high-margin lasers, we would expect it to increase its outsourcing of low-margin PDs - lasers (LDs) and PDs use the same equipment and capacity.
- We forecast a 91% revenue CAGR for VPEC's data-centre Photonics business over 2026/27E, and aiming for 60% segment GM in 2027E, resulting in overall gross margin rising 11pts over two years to hit ~48% by 2027E.

Figure 7 - Optical chip manufacturing process - VPEC's role



Source: Company, Macquarie Research, Nov 2025

Figure 8 - EML / CW laser structure



Microelectronics to sustain

- VPEC is a major supplier of GaAs-based epiwafers for power amplifiers (PA), a key component for wireless communication. The company claims over 70% market share in GaAs epiwafers.
- We believe impact from Skyworks'/Qorvo's merger to form an RF-focused semiconductor giant will be minimal, as both companies are already VPEC customers, and regulatory requirements mean the two companies will operate independently throughout 2026E.
- VPEC's gross margin is well above that of its competitors, with major rival IQE operating at a loss and planning to sell its Taiwan subsidiary, pointing to an easier competitive landscape ahead for VPEC.
- We take a conservative view on the smartphone market, but we still believe VPEC is well positioned in the wireless communications market, which includes 5G or future consumer products other than smartphones.

smartphone

Figure 10 - MQ smartphone shipment forecast

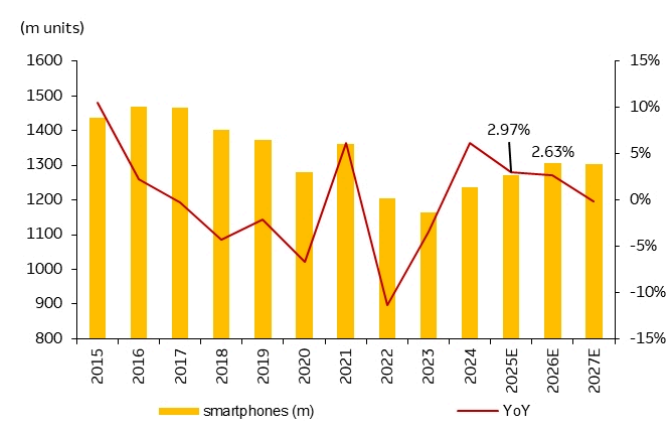
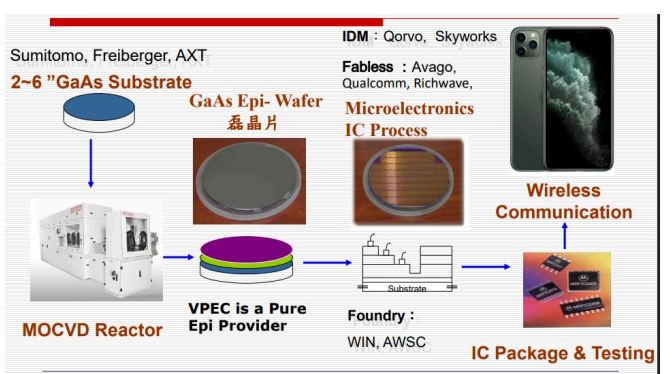


Figure 11 - VPEC's roles in RF industry



Source: Company, Macquarie Research, Nov 2025

VPEC produces epiwafers for two primary business lines:

Microelectronics - primarily for smartphone applications; and,

Photonics - focused on data centre usage

Company overview

Visual Photonics Epitaxy Co., Ltd. is a manufacturer of III-V compound semiconductor epitaxial wafers, with MOCVD as its core technology. The company's operations are divided into two segments:

1. **Microelectronics:** This segment focuses on GaAs epitaxial wafer products, which are mainly applied in smartphone PA and Wi-Fi radio frequency (RF) modules. VPEC holds the number one global market share in this segment.

2. Optoelectronics (Photonics): This segment supplies epitaxial wafers for data centre optical communications. Current products are more concentrated on PD epiwafers required for 400G and 800G specifications with lower contribution from LDs. The company is developing epiwafer products used in optical chips for 800G and 1.6T transceivers (including lasers) and expects major contribution in 2026E.

Q: What is an epiwafer?

An epiwafer (short for *epitaxial wafer*) is a semiconductor wafer that has one or more thin, high-purity crystalline layers grown on top of a base substrate (usually silicon, GaAs, InP, or GaN). It's a custom-engineered foundation for advanced chips — the “blank canvas” on which semiconductor devices are later fabricated. The epitaxial process allows precise control over the wafer's electrical properties (like doping, thickness, and defect density), which directly affects chip performance, power efficiency, and yield.

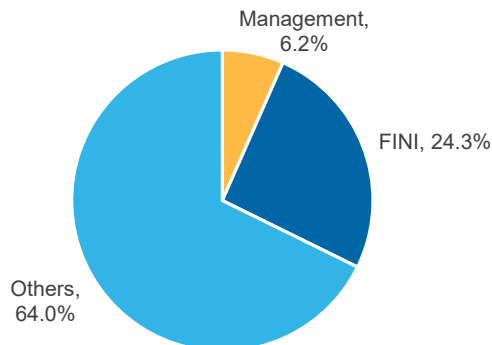
Epiwafers are critical for products such as power devices (SiC, GaN), RF chips, lasers, photodetectors, and optoelectronic components. Companies in this segment (e.g., VPEC, IQE, LandMark) act as upstream suppliers to foundries and device makers, enabling next-generation technologies in 5G, EVs, and optical communication in AI data centres.

Figure 12 - VPEC's epiwafer offering

Epi Material	Primary Use
GaAs	Radio Frequency Power Amplifier (in smart phones), VCSELs (for 3D sensing), Photodetectors (for datacom)
InP	CW lasers & Photodetectors (for data com)
GaN	Power Electronics (EV chargers), Blue LEDs

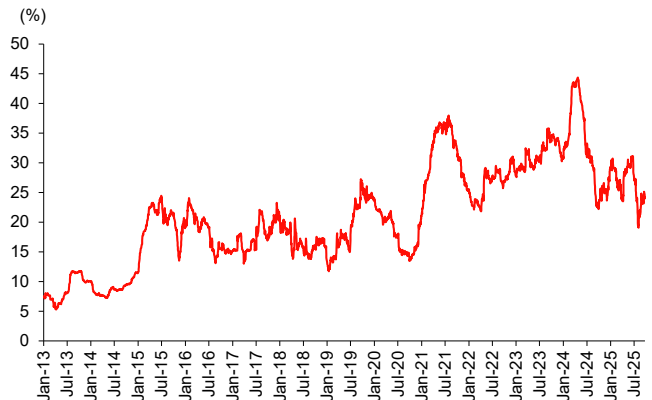
Source: Company, Macquarie Research, Nov 2025

Figure 13 - VPEC shareholding structure



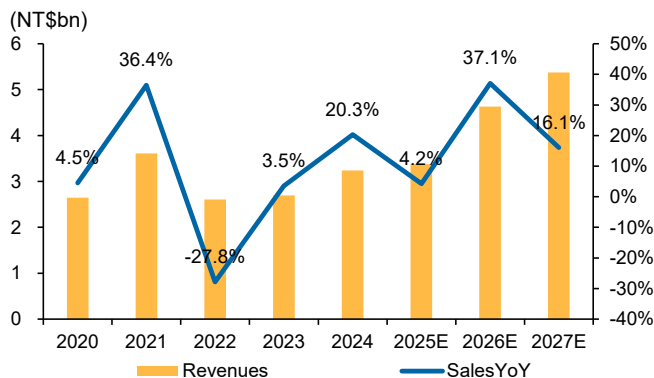
Source: TEJ, Macquarie Research, Nov 2025

Figure 14 - QFII holdings



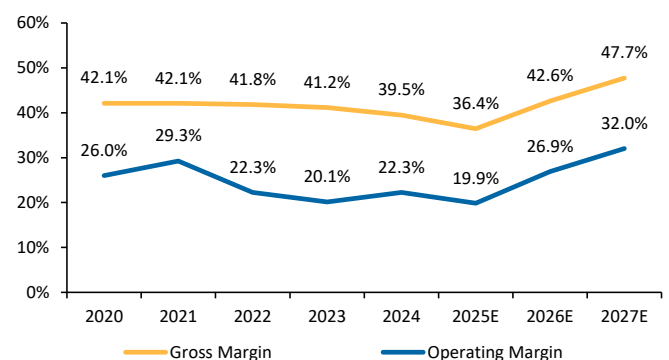
Source: TEJ, Macquarie Research, Nov 2025

Figure 15 - Sales / YoY growth



Source: Company, Macquarie Research, Nov 2025

Figure 16 - GM / OPM trend



Source: Company, Macquarie Research, Nov 2025

Figure 17 - P&L

NT\$m	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25E	2023	2024	2025E	2026E	2027E
Revenues	837	876	806	722	794	714	913	958	2,694	3,241	3,378	4,630	5,373
Gross Profit	341	347	331	260	317	246	300	369	1,109	1,279	1,231	1,972	2,565
Operating Profit	199	205	195	122	179	122	159	210	542	721	671	1,245	1,721
PBT	242	231	181	164	197	24	200	237	542	818	658	1,351	1,827
Net profit	193	201	144	133	157	36	160	190	450	671	544	1,081	1,462
EPS (NT\$)	1.04	1.09	0.78	0.72	0.85	0.20	0.86	1.02	2.43	3.63	2.93	5.81	7.86
Profitability													
Gross Margin	40.8%	39.6%	41.0%	36.0%	39.9%	34.4%	32.9%	38.5%	41.2%	39.5%	36.4%	42.6%	47.7%
EBIT Margin	23.8%	23.4%	24.2%	16.9%	22.6%	17.1%	17.4%	22.0%	20.1%	22.3%	19.9%	26.9%	32.0%
Pretax Margin	28.8%	26.4%	22.5%	22.7%	24.8%	3.4%	22.0%	24.7%	20.1%	25.2%	19.5%	29.2%	34.0%
Net Margin	23.0%	22.9%	17.9%	18.4%	19.8%	5.1%	17.6%	19.8%	16.7%	20.7%	16.1%	23.3%	27.2%
QoQ Growth													
Sales	-10.6%	4.6%	-8.0%	-10.3%	9.9%	-10.0%	27.8%	4.9%					
Operating Profit	-26.6%	2.9%	-5.0%	-37.3%	46.6%	-31.8%	30.4%	32.2%					
Pretax Profit	2.6%	-4.5%	-21.4%	-9.6%	19.9%	-87.6%	721.0%	18.2%					
Net Profit	2.4%	4.1%	-28.1%	-8.0%	18.4%	-76.9%	341.9%	18.1%					
EPS	2.4%	4.1%	-28.1%	-8.0%	18.4%	-76.8%	337.8%	18.1%					
YoY Growth													
Sales	112.3%	47.3%	4.9%	-22.9%	-5.2%	-18.4%	13.3%	32.6%	3.5%	20.3%	4.2%	37.1%	16.1%
Operating Profit	824.2%	115.1%	26.6%	-55.0%	-10.2%	-40.5%	-18.3%	72.3%	-6.5%	33.0%	-7.0%	85.6%	38.3%
Pretax Profit	1497.1%	107.0%	1.0%	-30.4%	-18.6%	-89.4%	10.5%	44.4%	-18.8%	50.9%	-19.5%	105.2%	35.3%
Net Profit	1542.2%	89.2%	0.3%	-29.5%	-18.5%	-81.9%	11.1%	42.6%	-17.3%	49.0%	-19.0%	98.8%	35.3%
EPS	1542.2%	89.2%	0.3%	-29.5%	-18.5%	-81.9%	10.5%	41.8%	-17.3%	49.0%	-19.3%	98.3%	35.3%

Source: Company, Macquarie Research, Nov 2025

Figure 18 - Balance Sheet

NT\$m	2019	2020	2021	2022	2023	2024	2025E	2026E	2027E
Cash & Cash Equivalents	875	1,066	1,210	753	826	1,176	1,032	1,033	2,347
Accounts Receivable	515	467	586	292	623	381	505	668	731
Inventories	330	372	415	487	505	619	788	955	965
Other Current Assets	65	67	91	88	92	108	140	140	140
TOTAL CURRENT ASSETS	1,785	1,972	2,302	1,620	2,045	2,284	2,464	2,795	4,183
Long-Term Investments	0	0	0	0	0	0	0	0	0
Gross Fixed Assets	5,333	5,392	5,498	6,120	6,387	6,430	6,510	6,910	7,310
Depreciation	-2,831	-3,114	-3,383	-3,617	-3,897	-4,183	-4,471	-4,771	-5,089
Construction in Progress	27	40	290	210	0	15	65	65	65
Net Fixed Assets	2,530	2,319	2,405	2,712	2,490	2,262	2,104	2,204	2,286
Other L/T Assets	25	28	143	80	42	38	118	115	112
TOTAL ASSETS	4,340	4,319	4,851	4,412	4,578	4,583	4,685	5,114	6,581
Short-Term Debt	470	380	480	200	100	0	0	0	0
Accounts Payable	304	317	372	176	397	337	429	520	525
Current Portion of Long-Term Debt	0	0	0	0	0	0	0	0	0
Other Current Liabilities	334	319	520	356	302	410	358	358	358
TOTAL CURRENT LIABILITIES	1,108	1,017	1,372	732	799	747	786	877	883
Long-Term Debt (ex. current portion)	400	400	200	590	700	500	600	600	600
Other Long Term Liabilities	1	0	0	7	9	7	4	4	4
TOTAL LIABILITIES	1,509	1,417	1,573	1,329	1,508	1,254	1,391	1,482	1,487
Common Stocks	1,849	1,849	1,849	1,849	1,849	1,849	1,849	1,849	1,849
Reserves	467	519	572	658	712	795	880	880	880
Retained Earnings	514	534	857	576	546	727	608	946	2,408
FX and Other Adjustments	0	0	0	0	-38	-42	-42	-42	-42
Treasury Stock	0	0	0	0	0	0	0	0	0
Minority Interest	0	0	0	0	0	0	0	0	0
SHAREHOLDERS' EQUITY	2,831	2,902	3,278	3,083	3,069	3,329	3,294	3,632	5,094
TOTAL LIABILITIES AND EQUITY	4,340	4,319	4,851	4,412	4,578	4,583	4,685	5,114	6,581

Source: Company, Macquarie Research, Nov 2025

Figure 19 - Cash flow statement

NT\$m	2019	2020	2021	2022	2023	2024	2025E	2026E	2027E
Net Income	514	533	855	545	450	671	544	1,081	1,462
Depreciation	255	284	271	242	282	286	288	300	318
Amortisation	1	1	1	1	2	2	3	3	3
Investment (Gain)/Losses - Equity	0	0	0	0	0	0	0	0	0
(Gain)/Losses from Disposals	0	0	0	0	0	0	0	0	0
Incr/(Decr) in Provisions	0	0	0	0	0	0	0	0	0
Net Change in Working Capital	-124	22	-109	26	-128	68	-187	-240	-68
Other	50	12	160	-171	-34	59	-76	0	0
Cash from Operations	695	851	1,179	644	572	1,086	572	1,144	1,714
Capital Expenditures	-193	-95	-467	-418	-68	-44	-199	-400	-400
Acquisition of L/T Investments	0	0	0	0	0	0	0	0	0
Proceeds from Disposal of S/T Investments	0	0	0	0	0	0	0	0	0
Other	-1	-2	-3	-52	-3	-3	-2	0	0
Cash from Investing	-195	-97	-470	-470	-71	-47	-201	-400	-400
Net Change in LT Debt	20	1,500	1,280	1,630	2,590	1,900	600	0	0
Net Change in ST Debt	-30	-90	100	-280	-100	-100	0	0	0
Dividends Paid	-462	-462	-479	-740	-425	-407	-592	-742	-744
Proceeds from Share Issuance	0	0	0	0	0	0	0	0	0
Employees & Director Bonus	0	0	0	0	0	0	0	0	0
Other Financing Cash Flows	240	-1,502	-1,482	-1,242	-2,483	-2,104	-503	0	0
Cash from Financing	-232	-554	-581	-632	-419	-711	-495	-742	-744
Currency Adjustments	-10	-8	16	1	-9	22	-20	0	0
Net Cash Flow	258	192	144	-457	73	350	-144	1	570
Free Cash Flow	502	756	711	226	504	1,042	374	744	1,314

Source: Company, Macquarie Research, Nov 2025

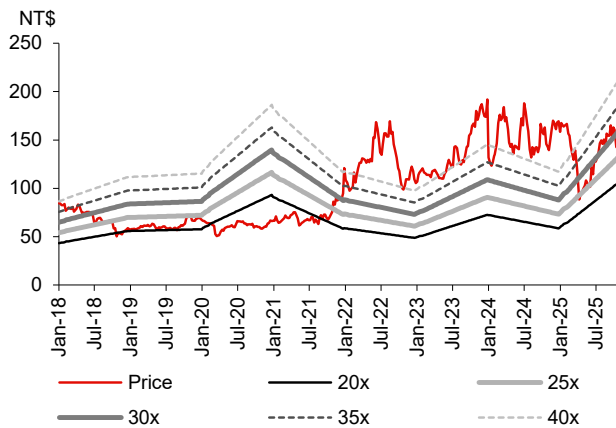
Valuation

- Our target price NT\$236 is based on 30x 2027PE. VPEC has been trading at a premium range of above 30x on average since 2018. It is mainly due to its stable GM from its microelectronic business and its leadership in the epiwafer industry.
- We believe the 30x can be justified by 2025-27E EPS CAGR at 64%, driven by larger revenue from CW lasers and higher mix of margin-accretive Photonics business.
- Taiwanese peers are currently trading at 25-45x forward P/E multiples (ignoring extreme values for FOCl and GCS given low earnings bases) due to 1) proximity to the world's largest AI hardware supply chain; 2) CPO's potential as a solution, which would circumvent geopolitical risk associated with China's dominance of the transceiver market, as CPO technology is less China-reliant and more dependent on Taiwan's semiconductor technology, and 3) rapid pace of revenue growth due to low base.
- Our target multiple remains below that of some of its peers, as we believe the market has yet to view VPEC as an optical chip player; as such, we see potential for a stronger rerating beyond our target multiple as the market recognises its changing nature.

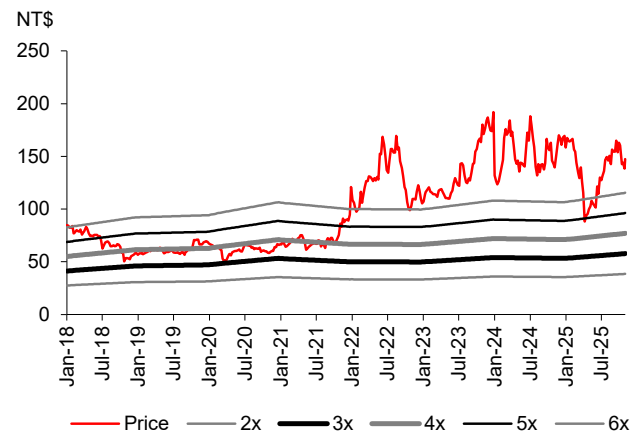
Figure 20 - Taiwanese optical communication names

			Mkt Cap	CP	TP	TSR	EPS (LC)			PER (X)			P/BV (X)			ROE (%)			Div yield (%)		
Ticker	Company	Rating	(US\$m)	(LC)	(LC)	(%)	25E	26E	27E	25E	26E	27E	25E	26E	27E	25E	26E	27E	25E	26E	27E
Taiwan optical communication names																					
3081 TT	Landmark	O	1,222	410	480	17%	4.0	10.0	13.7	103.7	41.0	29.9	9.1	7.5	6.1	8.8	18.4	20.4	0.1	0.2	1.0
2455 TT	VPEC	O	888	149	236	64%	2.9	5.8	7.9	50.9	25.7	19.0	8.4	7.6	5.4	16.5	29.7	28.7	2.7	2.7	2.7
3363 TT	FOCI	NR	1,220	358	NA	NA	-0.5	0.8	2.7	n.a.	458.3	131.9	14.6	14.7	13.7	-2.1	3.2	10.8	0.0	0.2	0.3
4991 TT	GCS	NR	611	164	NA	NA	-2.1	0.2	5.4	n.a.	883.8	30.4	6.1	6.1	5.0	-7.9	-0.3	14.6	0.0	na	na
3450 TT	Elaser	NR	1,188	257	NA	NA	3.8	6.4	10.6	67.1	40.2	24.1	8.4	7.7	6.6	13.5	21.3	34.6	0.2	1.8	3.1
6442 TT	EZConn	NR	2,587	947	NA	NA	14.3	19.3	31.3	66.2	49.0	30.3	19.5	14.7	10.6	36.8	30.8	41.1	0.9	0.8	1.3

Source: Bloomberg, Macquarie Research, Nov 2025. Close prices as of 12 November

Figure 21 - 12-month forward P/E band

Source: TEJ, Macquarie Research, Nov 2025

Figure 22 - 12-month forward P/B band

Source: TEJ, Macquarie Research, Nov 2025

Risks to our investment thesis

- **Slower-than-expected transceiver demand:** A delay in Nvidia's next-gen GPU server might delay the adoption of AI networking equipments, with knock-on impacts on demand for optical chips and epiwafers.
- **Excessive capacity at leading players:** Optical chips demand has been strong since 2024 and capacity expansion plans are on-going for leading players. Accelerated expansion plans across the industry could result in substantial oversupply and price erosion, a phenomenon with ample historical precedent.
- **Declining wireless communication market:** A slower smartphone market will impact on VPEC's microelectronic business as it boasts 70% of market share in epiwafers for RF chips, leading to potential sales decline and margin erosion.

LandMark (3081 TT, OP, TP: NT\$480 - PER)

- **Slower-than-expected transceiver demand:** A delay in Nvidia's next-gen GPU server might delay the adoption of AI networking equipments, thus potentially reducing our expected growth in demand for products such as CW lasers and the outsourcing PD market.
- **Excessive capacity at leading players:** Optical chips demand has been strong since 2024 and capacity expansion plans are on-going for leading players. Accelerated expansion plans across the industry could result in substantial oversupply and price erosion, a phenomenon with ample historical precedent.
- **Failed or delayed 3" qualification:** LandMark only has 2" capacity before the InP substrate ban, and CSP customers requires extensive testing (~6,000 hours for lasers before any product transition.) Failure or delays in securing qualification could lead to orders shift away from LandMark to other CW laser vendors, as customers continue to face pressure from demanding CSP and AI GPU customers, with their intense deployment roadmaps.

Key Quant Findings

The quant model currently holds a reasonably negative view on Visual Photonics Epitaxy. The strongest style exposure is Quality, indicating this stock is likely to have a superior and more stable underlying earnings stream. The weakest style exposure is Earnings Momentum, indicating this stock has received earnings downgrades and is not well liked by sell side analysts.

Macquarie Alpha Model: Key rankings

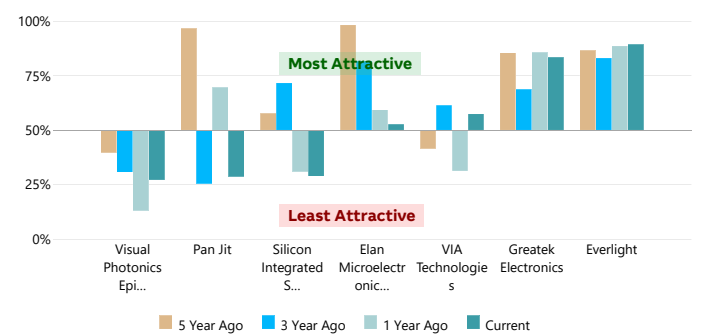
The Macquarie Quant’s flagship Alpha model is a dynamic multi-factor model based on a staple of quant factors such as value, momentum, revisions, quality, and risk.

	Global	Market (Country)	Sector
	Whole Universe	Taiwan	Semiconductors & Semiconductor Equip.
Macquarie Alpha Model	13416/17935	714/979	425/650
Fundamental (Consensus) *	3247/17935	63/979	138/650

* based on Total Shareholder Return = Consensus Price target / Current Price

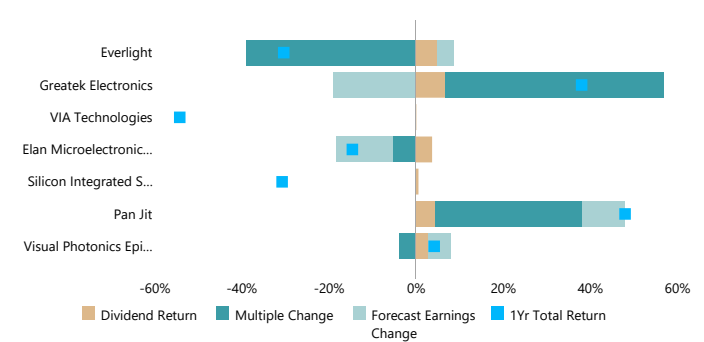
Current and Historical Alpha Model Rank

The chart shows the Macquarie Alpha model market ranking against the company's peers and over recent history.



Drivers of Stock Return

Breakdown of 1-year total return (local currency) into returns from dividends, changes in forward earnings estimates and the resulting change in earnings multiple.



Source (all charts): FactSet, Refinitiv, and Macquarie Quant. For more details on the Macquarie Alpha model or for more customised analysis and screens, please contact the Macquarie Global Quantitative Team: maccapequitiesresearchquantglobal@macquarie.com. Explanation for items on this page can be found at <https://www.macquarieinsights.com/rp/dr/rp/OTUyMzg1>

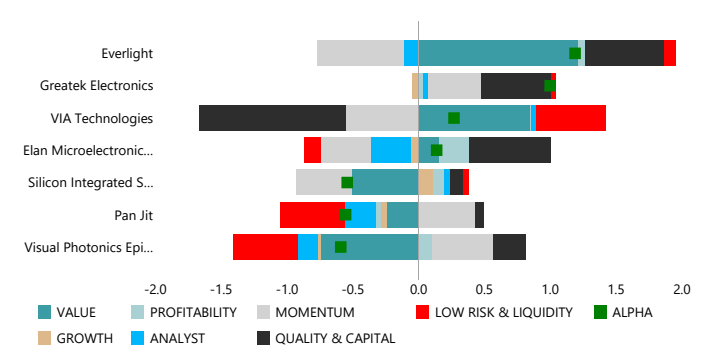
Alpha Model Decomposition

The Macquarie Alpha is decomposed into its sector and market relative factor & styles exposures (a higher/better percentile is coded in green, whilst lower in red).

Factors / Styles	Percentile relative to		Core factors in definition
	sectors (/650)	market (/979)	
ALPHA	35%	27%	Built from the styles below
VALUE	49%	11%	Book, CF, Yield, Earnings Multiples
ANALYST	44%	68%	Revisions (Earnings, Recommendations)
MOMENTUM	60%	82%	Price Momentum
GROWTH	35%	27%	EPS, Sales (Forecast, Historic)
PROFITABILITY	75%	88%	ROE, Margin, Asset Turnover
QUALITY	75%	89%	Accruals, Earn Stability, Cash Conversion
CAPITAL	76%	52%	Investment/Capex, Net share issuance
LIQUIDITY	19%	27%	Size, Turnover, Analyst Coverage
LOW RISK	23%	16%	Beta, Volatility, Earn.Cert, Leverage
TECHNICAL	8%	10%	MACD, RSI, Bollinger, Williams R, etc

Factors driving the Alpha Model vs peers

For the comparable firms this chart shows the key underlying styles and their contribution to the current overall raw Alpha score.



Macquarie Style Returns over last year

Recent performance to Macquarie style factors

Asia Ex JP	Monthly Factor Long-Short Returns for												Last 5 Years (ann)	Last 10 Years (ann)
	Oct - 25	Sep - 25	Aug - 25	Jul - 25	Jun - 25	May - 25	Apr - 25	Mar - 25	Feb - 25	Jan - 25	Dec - 24	Nov - 24		
ALPHA	2%	-3%	-6%	-7%	-1%	1%	0%	4%	-3%	1%	7%	1%	3%	4%
VALUE	4%	-4%	-4%	-3%	2%	4%	-2%	4%	-3%	1%	5%	1%	9%	4%
ANALYST	3%	4%	1%	-3%	4%	0%	-1%	1%	2%	3%	6%	0%	13%	11%
MOMENTUM	-2%	0%	-6%	-6%	-2%	2%	0%	-2%	-1%	-1%	4%	2%	-5%	0%
GROWTH	0%	5%	8%	0%	1%	0%	-1%	-1%	0%	0%	5%	2%	-1%	1%
PROFITABILITY	0%	-4%	-2%	-6%	1%	3%	1%	5%	-5%	1%	4%	0%	-7%	-3%
QUALITY	-4%	-6%	1%	-3%	-4%	0%	1%	2%	-4%	1%	4%	1%	-7%	-2%
CAPITAL	-3%	-1%	-2%	-1%	-1%	0%	0%	3%	-4%	-1%	1%	1%	1%	-3%
LIQUIDITY	-1%	3%	3%	-2%	1%	2%	0%	-1%	0%	0%	-1%	2%	-5%	1%
LOW RISK	1%	-6%	-11%	-9%	-5%	-1%	4%	5%	-7%	-1%	4%	2%	-8%	-6%

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Important Disclosures

Recommendation definitions	Volatility index definition	Financial definitions
Macquarie – Asia and USA Outperform – expected return >10% Neutral – expected return from -10% to +10% Underperform – expected return <-10% Macquarie – Australia/New Zealand Outperform – expected return >10% Neutral – expected return from 0% to 10% Underperform – expected return <0% During periods of share price volatility, recommendations and target prices may occasionally and temporarily be inconsistent with the above definitions. Recommendations – 12 months 12-month target – Expected share price in 12 months Valuation – The company's estimated fair value share price based on the disclosed valuation methodology Note: Quant recommendations may differ from Fundamental Analyst recommendations	This is calculated from the volatility of historical price movements. Very high – highest risk – Stock should be expected to move up or down 60-100% in a year – investors should be aware this stock is highly speculative. High – stock should be expected to move up or down at least 40-60% in a year – investors should be aware this stock could be speculative. Medium – stock should be expected to move up or down at least 25-40% in a year. Low – stock should be expected to move up or down at least 15-25% in a year. * Applicable to select stocks in Asia/Australia/NZ Note: expected return is reflective of a Medium Volatility stock and should be assumed to adjust proportionately with volatility risk	All "Adjusted" data items have had the following adjustments made: Added back: goodwill amortisation, provision for catastrophe reserves, IFRS derivatives & hedging, IFRS impairments & IFRS interest expense Excluded: non recurring items, asset revals, property revals, appraisal value uplift, preference dividends & minority interests EPS = adjusted net profit / efpowa* ROA = adjusted ebit / average total assets ROA Banks/Insurance = adjusted net profit / average total assets ROE = adjusted net profit / average shareholders funds Gross cashflow = adjusted net profit + depreciation *equivalent fully paid ordinary weighted average number of shares All Reported numbers for Australian/NZ listed stocks are modelled under IFRS (International Financial Reporting Standards).

Recommendation proportions for quarter ending 30 Sept 2025

	AU/NZ	Asia	USA	
Outperform	51.96%	67.82%	67.16%	(for global coverage by Macquarie, 1.45% of stocks followed are investment banking clients)
Neutral	38.79%	18.36%	32.84%	(for global coverage by Macquarie, 1.75% of stocks followed are investment banking clients)
Underperform	9.25%	13.83%	0.00%	(for global coverage by Macquarie, 0.00% of stocks followed are investment banking clients)

Company-Specific Disclosures

Company Name	Disclosure
Visual Photonics Epitaxy (2455 TT) Outperform 12-month target: TWD236.00 - PER Valuation: TWD 236.00 - PER Price: TWD156.50	None
LandMark Optoelectronics (3081 TT) Valuation: TWD 480.00 - PER	None

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